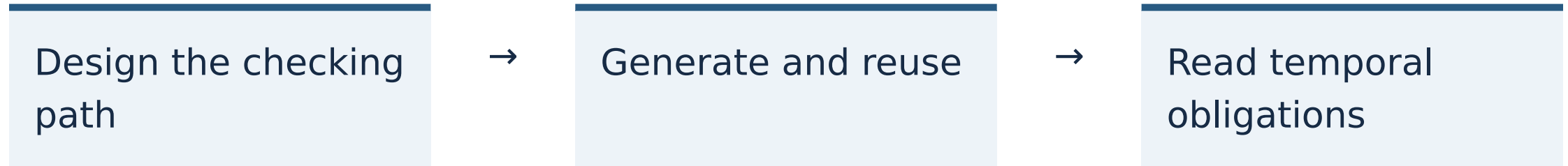


Dynamic verification

Every technique must discharge a named verification claim.



Learning outcomes

Demonstrate these decisions on a real plan row.

- Choose a scoreboard identity without losing an ordering check.
- Separate stimulus legality from distribution.
- Read assertion activation, timing and reset.
- Define a reproducible regression and triage policy.

Route through the lesson

Four blocks, each ending with a decision.

- Design the checking path
- Generate and reuse
- Read temporal obligations
- Operate the regression

Separate environment responsibilities

Observation and measurement must not alter behavior.

- Keep stimulus, driving and monitoring distinct.
- Keep checking independent of coverage collection.

The scoreboard key is an engineering decision

Reordering tolerance can erase an ordering check.

- Choose a key from observable identity.
- Account for independent concurrency sources.
- Retain explicit ordering obligations.

A predictor can be wrong

Golden applies to a specification, not to confidence in code.

- Review the reference model's origin.
- Compare at a compatible abstraction boundary.

Discussion: tolerate reordering?

Name the evidence that would settle the decision.

- A content-keyed scoreboard passes reordered responses.
- The interface requires ordering within a stream.
- What check has been lost?

Legality and distribution differ

A solvable model can still exclude the intended scenario.

- Review the legal set separately from probability.
- Measure the generated distribution.

A seed is not a complete reproducer

Execution context belongs in the run record.

- Record source revision, tool build and configuration.
- Preserve random-stability assumptions.

UVM supplies reusable structure

It does not supply the plan, checker or coverage intent.

- The interface agent is the reuse boundary.
- A register mirror is not a functional scoreboard.

Discussion: the framework is complete

Name the evidence that would settle the decision.

- The UVM topology, sequences and register layer run.
- The feature has no pass/fail criterion.
- What is still missing?

An assertion is a sampled statement

Clocking and reset define what is evaluated.

- Identify the sampling event.
- Separate same-tick and future obligations.
- State how reset discards attempts.

Stable can look backward

A plausible operator can express the wrong time relation.

- A fresh request may legally introduce a new value.
- The hold obligation concerns the following sampled tick.

Cover the antecedent

A clean implication can be unexercised.

- Distinguish satisfaction from vacuity.
- Use a witness for the condition that starts the check.

Discussion: all assertions pass

Name the evidence that would settle the decision.

- The traffic never stalls.
- A hold-while-stalled property never fails.
- Which witness should the review demand?

Tiers are admission policies

Runtime budgets belong to suite design.

- Place fast static checks before costly runs.
- Give each tier a purpose and entry rule.

Triage before deep debugging

Failure signatures must preserve useful distinctions.

- Reduce duplicate failures.
- Separate design, environment and infrastructure causes.
- Keep a reproducible representative.

A ranked suite does not explore new behavior

Selection needs an independent exploration budget.

- Investigate flakes instead of training teams to ignore them.
- Keep unranked exploration alongside selection.

Discussion: rerun until green

Name the evidence that would settle the decision.

- An intermittent failure passes on rerun.
- The dashboard replaces the original result.
- What evidence was discarded?

Exercise: choose the revealing antecedent

Equivalent formulas can expose different vacuity.

- Compare $ok \implies \text{not err}$ with $err \implies \text{not ok}$.
- Assume err is rare and deliberately constructed.
- Choose a formulation and the necessary activation witness.

What to change on Monday

Make one existing verification decision reviewable.

- Audit one scoreboard key.
- Reproduce a run from its recorded manifest.
- Find an assertion without an activation witness.

Scope and next reading

Use the handout for the detailed artifacts and sources.

- Read the Dynamic verification handout and book Chapters 8–12.
- Detailed UVM APIs and complete environment implementation remain in the handout.